

# Foundations Notebook

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# Chapter 1

## Mathematical Foundations

### 1.1 Linear Algebra

#### 1.1.1 Vector Spaces

**Definition 1.1.1.** A vector space  $V$  over a field  $\mathbb{F}$  is ...

#### 1.1.2 Why This Matters

Linear algebra underpins signal spaces, optimization, and estimation theory.

## Chapter 2

# Electrical Engineering Foundations

### 2.1 Electro Magnetism

### 2.2 Radar

Radar: Radio Detection And Ranging. Is simply the transmission and receiving of Radio waves. From these radio waves it is possible to determine many things about the incident target.

#### Simple Model

A transmitter is to be thought as a isotropic radiator with the propogation pattern described by equation 2.1.

$$P_d = \frac{P_t}{4\pi r^2} \quad (2.1)$$

Where  $P_d$  is the power desnity at some radius,  $P_t$  is the power transmitted and  $r$  is the radius.

$$R = \frac{c\delta t}{2} \quad (2.2)$$

### **2.2.1 SAR**

## **2.3 Signals and Systems**

### **2.3.1 Sampling**

### **2.3.2 LTI Systems**

An LTI system satisfies:

- Linearity
- Time invariance

### **2.3.3 Idea**

- Optical PCB's

# Chapter 3

## Leadership

### 3.1 Objective

As a military officer my job is to manage the application of violence

### 3.2 Culture

Between branches the ideals revolve around the unit of action. In the case of the navy the unit of action is the ship, the army is the squad, and the airforce the plane. The ability for the unit of action to execute its mission is what drives the culture. In the case of the navy, for a ship to execute its mission the entire ship crew and all must act as a unified machine. Little deviation can be accepted as to ensure proper function of the ship. Hence the navy has a culture of detail driven instruction and execution. While the army being based around the squad needs to rely on the flexibility and innovation of the individuals in the squad to maintain effectiveness. Thus the army may focus more on the presentation and management of an officer as opposed to the navy's greater focus on technical ability.

### 3.3 Ideals

1. Give a shit
2. Know my shit
3. Don't be a shit
4. Get shit done